

Correlation & Regression Exercise

Methods in Computational Linguistics

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Introduction

The main goal of this lab session is to get a feeling for correlations, regression, and evaluation. We will start with calculating a correlation coefficient, use it to calculate a regression, and then convert this task into a classification problem that we evaluate with the accuracy metric.

The participation in this lab session is voluntary: you do not have to submit any result. The work in this session should be done individually, but you are encouraged to discuss with your colleagues. As always, there is not only one correct implementation for this exercise. You can but do not have to use the suggested structure in the python file on Ilias. You also can - and are encouraged to - use your own data. If you want to get feedback, please ask questions during the session.

Example Data

We want to predict the (continuous) toxicity score of a text (y) given the number of profound words in it (x). We use some synthetic example data:

$$\begin{aligned}x_{train} &= [7, 3, 3, 0, 0, 1, 4, 2, 0, 1] \\y_{train} &= [1.00, 0.39, 0.00, 0.00, 0.25, 0.00, 0.27, 0.17, 0.38, 0.12] \\x_{test} &= [1, 5, 0, 0, 6, 2, 0, 1, 2, 4] \\y_{test} &= [0.62, 0.45, 0.08, 0.55, 0.88, 0.30, 0.12, 0.20, 0.58, 0.43]\end{aligned}$$

If not stated otherwise, use the train subset of the data.

Correlation

Calculate the Pearson Correlation Coefficient r between x and y for n datapoints.

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

What is your interpretation of the result?

Regression

Now, train a simple linear regression model by calculating the slope b and the intercept a using the formula from the lecture.

$$\begin{aligned}b &= r \frac{s_y}{s_x} \\a &= \bar{y} - b\bar{x}\end{aligned}$$

Where r is the Pearson correlation coefficient and s_x and s_y are the standard deviations of x and y . You can calculate them with the following formula:

$$s_x = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

After calculating the parameters, write a prediction function using the regression equation.

R^2 Statistic

We now want to evaluate the model fit on the training data. For that, you should calculate the R-squared score of the regression model:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

What is your interpretation of the score?

Evaluation

Now, convert the regression task into a very simple classification task that we will evaluate [on our test data](#). First, predict the toxicity score using our regression function. Second, convert this score into a binary label using the following conditions:

$$\hat{y}_{binary} = \begin{cases} 1 & \text{if } \hat{y} \geq 0.5 \\ 0 & \text{if } \hat{y} < 0.5 \end{cases}$$

Now calculate the accuracy and the F1-score of the prediction. What is your interpretation of the result?